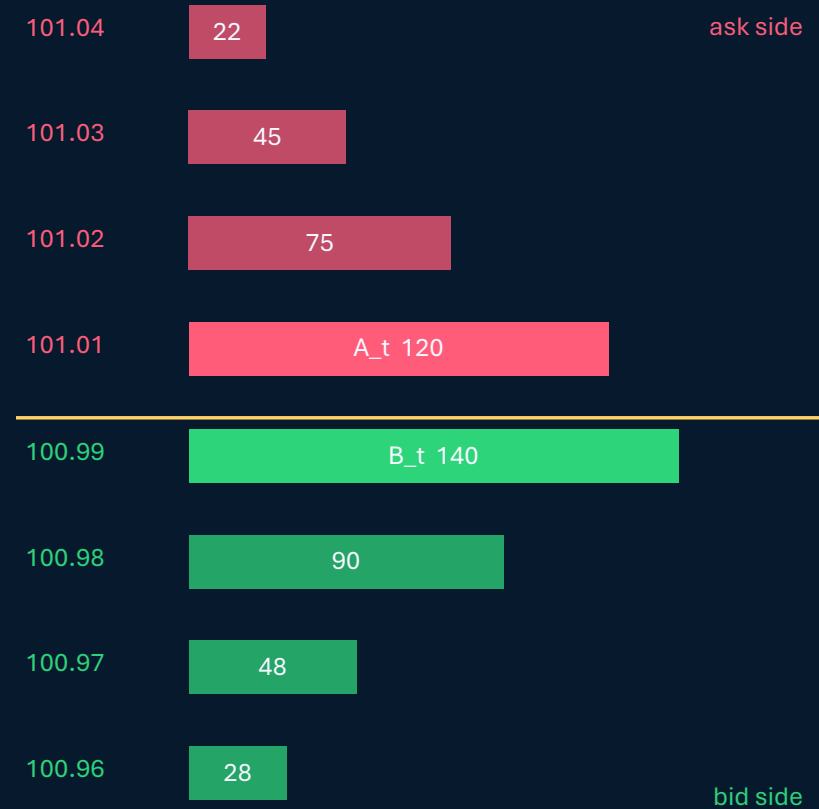


Deep Learning for Limit Order Books

Neural and Spatial Models for Short-Term LOB
Dynamics

Ids Rehorst
WI4224: Special Topics in Financial Engineering
TU Delft



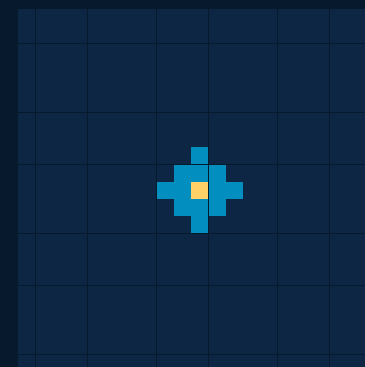
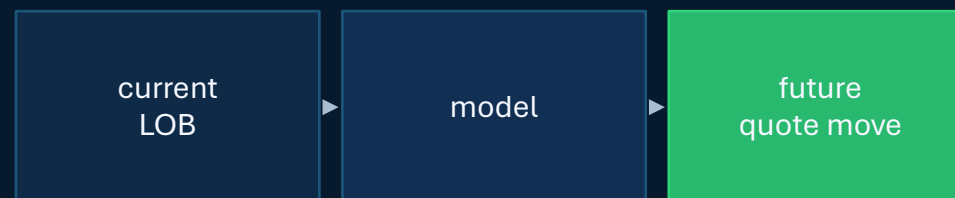
Question

Motivation

Can the current limit order book state predict short-term quote movements?

$$\mathbb{P}((\Delta A_t, \Delta B_t) | X_t)$$

- X_t = current limit order book state
- A_t = best ask
- B_t = best bid

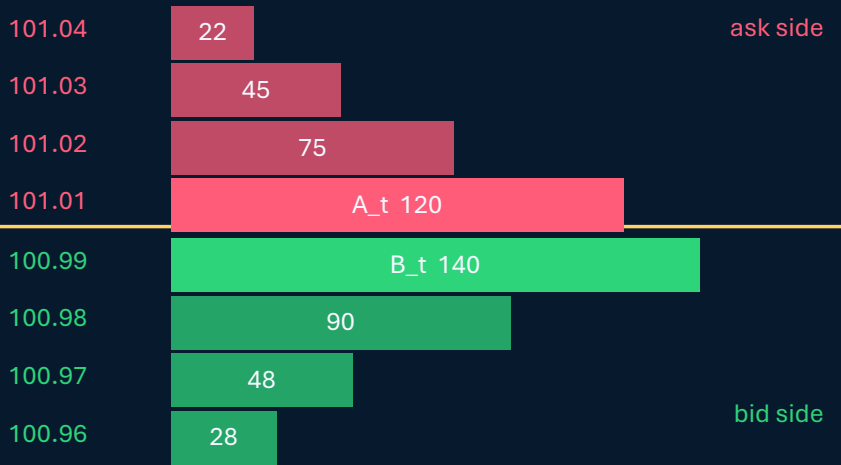
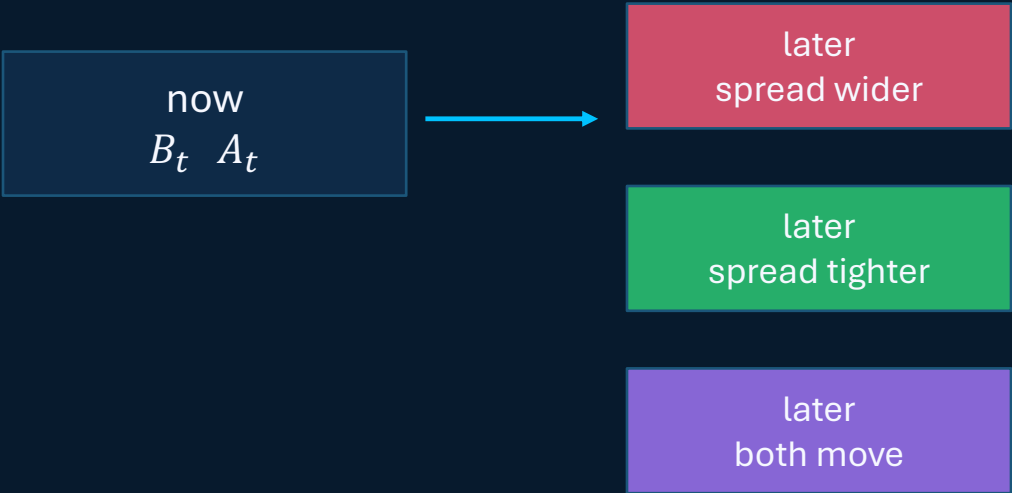


Output: probability distribution over joint quote movements

- market making
- execution algorithms
- short-term risk management
- market microstructure research

A trader does not only care whether the mid-price moves.

They care how the best bid and best ask move separately.



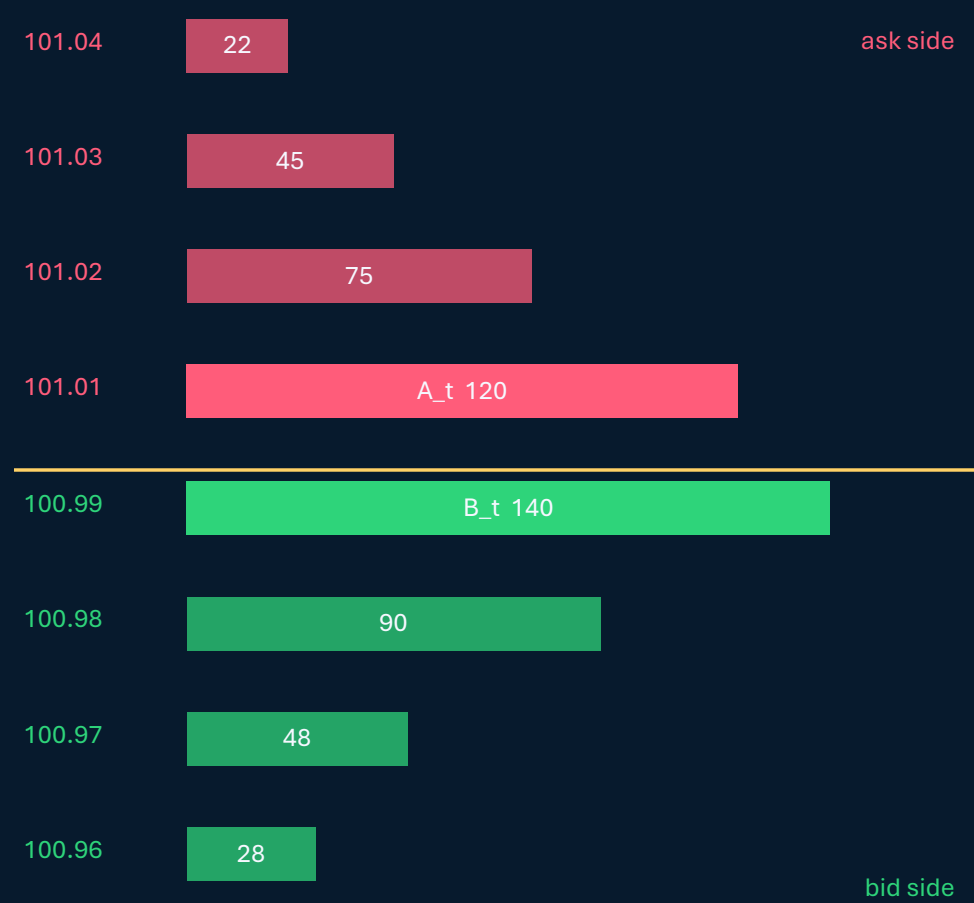
B_t = best bid = highest buy price

A_t = best ask = lowest sell price

$S_t = A_t - B_t$

$M_t = (A_t + B_t)/2$

Normally: $B_t < A_t$

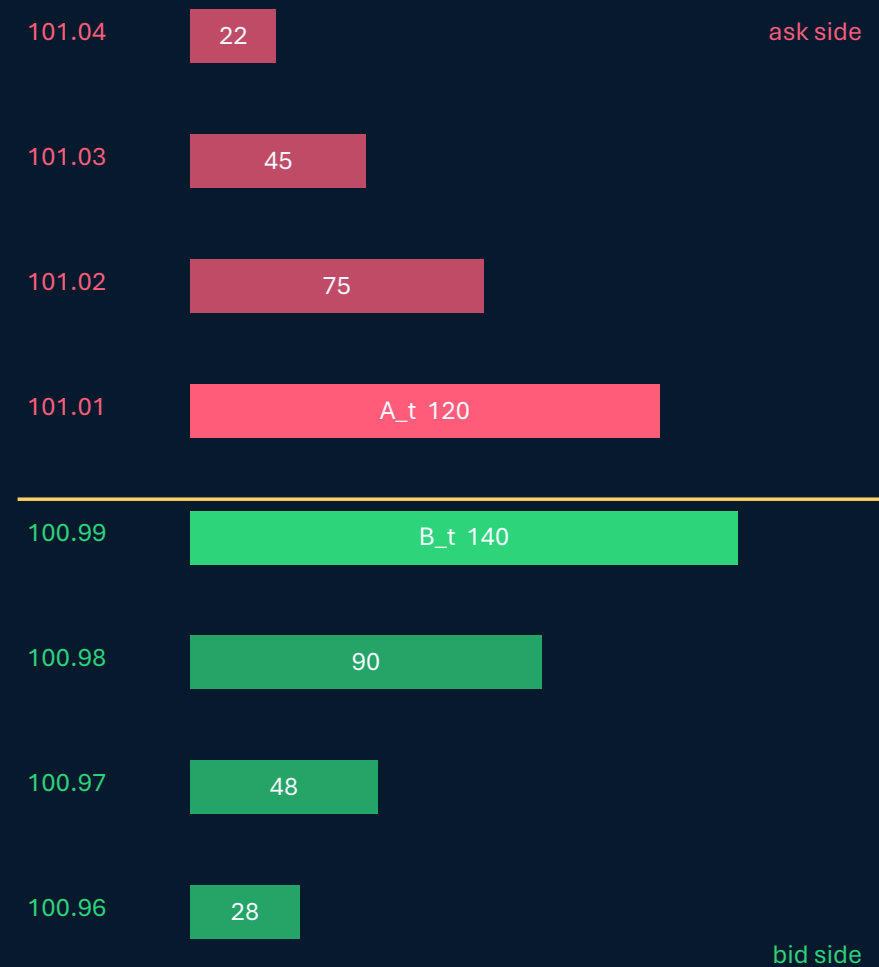


We observe L levels on both sides.

Ask: $(p_t^{a,1}, v_t^{a,1}), \dots, (p_t^{a,L}, v_t^{a,L})$

Bid: $(p_t^{b,1}, v_t^{b,1}), \dots, (p_t^{b,L}, v_t^{b,L})$

- p = price level
- v = available volume
- L = number of observed levels



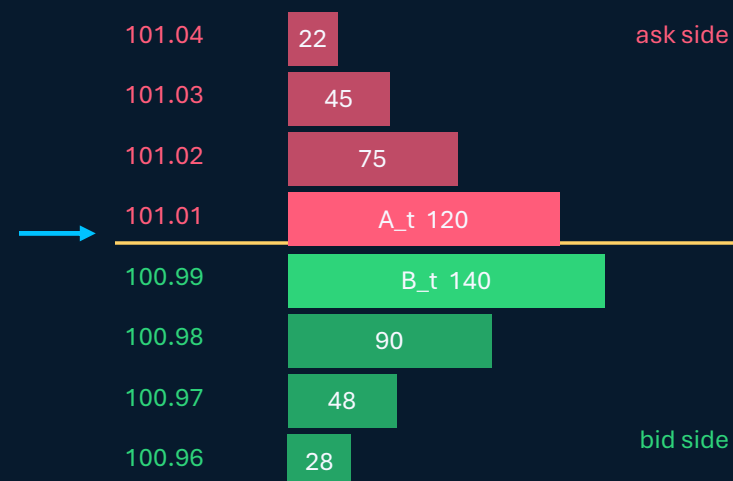
The book state is represented by X_t .



$$X_t \in R^p$$

In the final experiments:

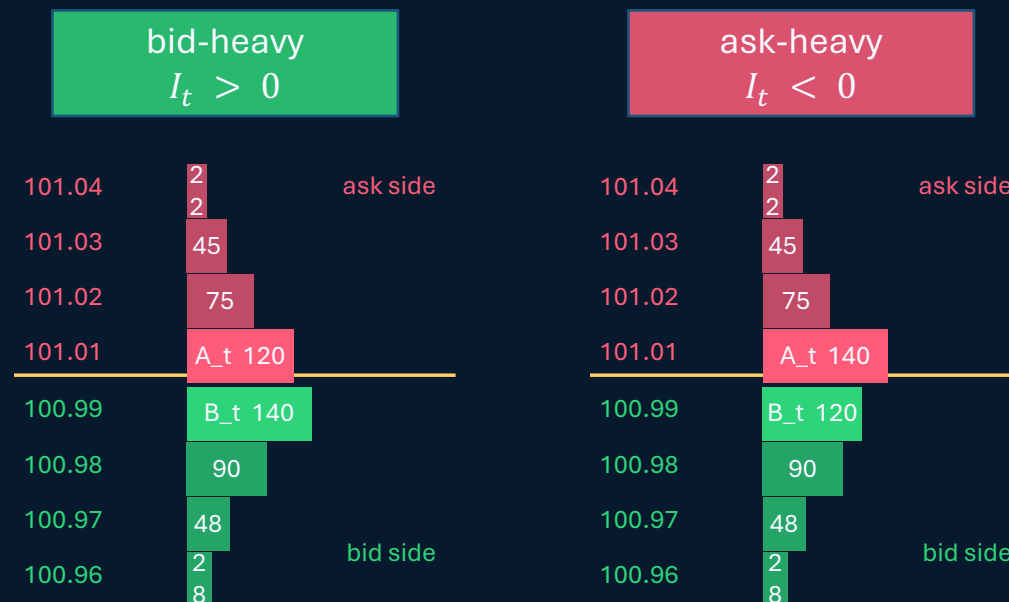
L = 50 levels
features: volumes, offsets, spread,
imbalance



Imbalance

$$I_t = (v_t^{b,1} - v_t^{a,1}) / (v_t^{b,1} + v_t^{a,1})$$

- $I_t > 0$: more volume at the best bid
- $I_t < 0$: more volume at the best ask
- computed level-by-level in the implementation



Imbalance summarizes pressure between the two sides of the book.

Target

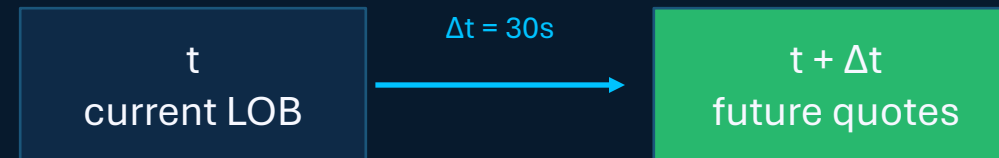
Problem

$$Y_t = (\Delta A_t, \Delta B_t)$$

$$\Delta A_t = (A_{t+\Delta t} - A_t) / \text{tick size}$$

$$\Delta B_t = (B_{t+\Delta t} - B_t) / \text{tick size}$$

Y_t measures future quote movement
in ticks.



$$(A_t, B_t) \rightarrow (A_{t+\Delta t}, B_{t+\Delta t})$$

Classes

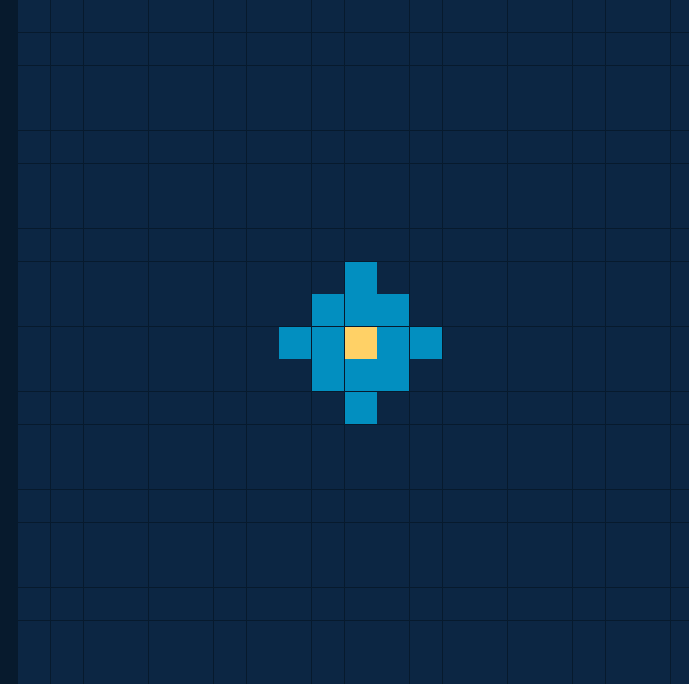
Problem

$$Y = \{-K, \dots, K\}^2$$

In our experiments:

$$K = 10$$

$$(2K + 1)^2 = 21^2 = 441 \text{ classes}$$



center: no movement (0,0)

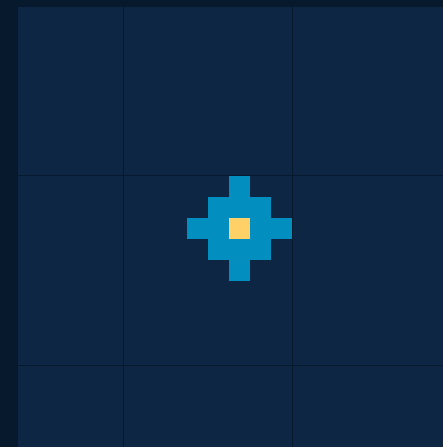
Goal

Problem

estimate $\mathbb{P}(Y_t = y | X_t), y \in \{-K, \dots, K\}^2$

This is not only classification.

The model outputs a full probability distribution.



probability heatmap over
future quote movements

estimate $\mathbb{P}(Y_t = y | X_t), y \in \{-K, \dots, K\}^2$

Classical LOB models

Queueing / stochastic models of order book dynamics

Useful for intuition, but often simplified and hard to use for fast full-distribution prediction.

Deep learning for LOBs

CNNs, LSTMs, MLPs and order-flow models for short-term price prediction

Often focus on mid-price direction or returns, not the joint bid/ask distribution.

Sirignano (2016)

Spatial neural network for the joint distribution of future best bid and ask prices

Key idea: exploit local structure across price levels deep in the book.

estimate $\mathbb{P}(Y_t = y | X_t), y \in \{-K, \dots, K\}^2$

Sirignano:

full conditional distribution of future best bid/ask movements

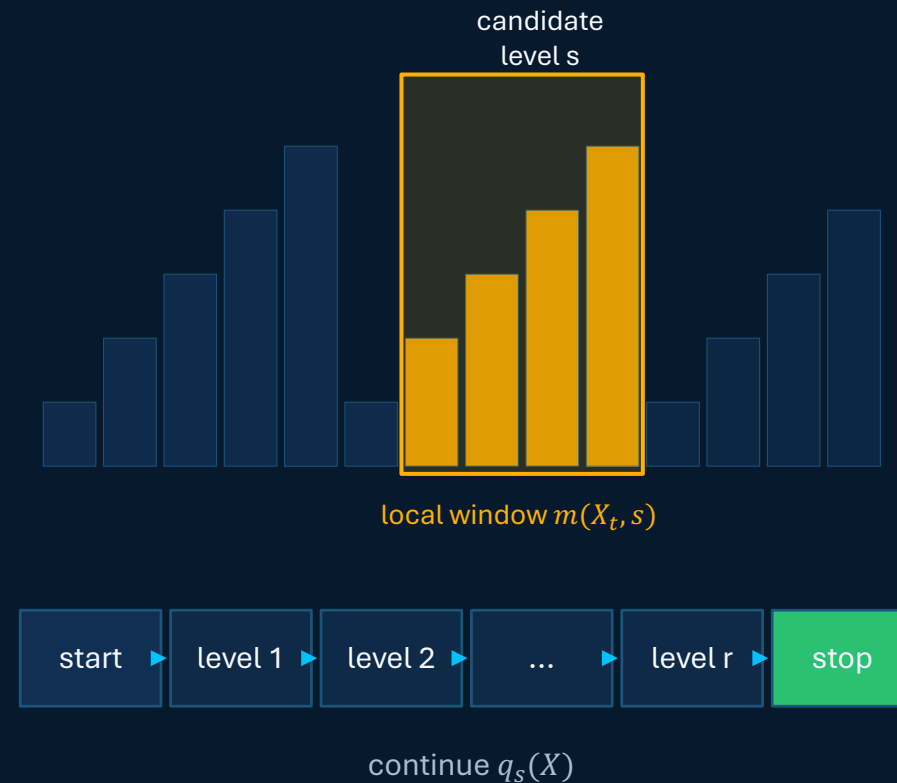
Kolm, Turiel & Westray:

multi-horizon return / alpha forecasting using LOB and order-flow features

Starting point: Sirignano Deep Learning for Limit Order Books

- model joint best ask / best bid movements
- compare simple baselines and neural networks
- exploit spatial structure in price levels

Original finding:
spatial neural networks strongly
outperform standard neural networks.



Original paper

489 stocks

NASDAQ Level III

~50 TB raw data

large GPU setup

full spatial NN

Challenges:
Data availability
Scale

Possible data sources

Lobster data free samples
FI-2010 dataset
Ethereum LOB data
Generate LOB data

What do we need?

Dataset large enough to train
neural network
50 LOB levels
Free

WSELOB-2017

- five Warsaw Stock Exchange stocks
- full year of order book data
- roughly 10 million usable samples per stock

Stock	Usable samples
KGHM	10.03M
PEKAO	10.12M
PKNORLEN	10.11M
PKOBP	9.84M
PZU	9.98M

Large enough for neural models,
but still very different from the original
NASDAQ setting.

Original paper	This project
489 stocks	5 stocks
NASDAQ Level III	WSELOB-2017
~50 TB raw data	public HDF5 data
large GPU setup	local-machine feasible
full spatial NN	reduced + full-ish spatial

Core question:
Does the spatial idea still help
on a smaller public dataset?

Raw WSELOB events are converted into fixed-frequency supervised learning samples.

50
levels

30s
horizon

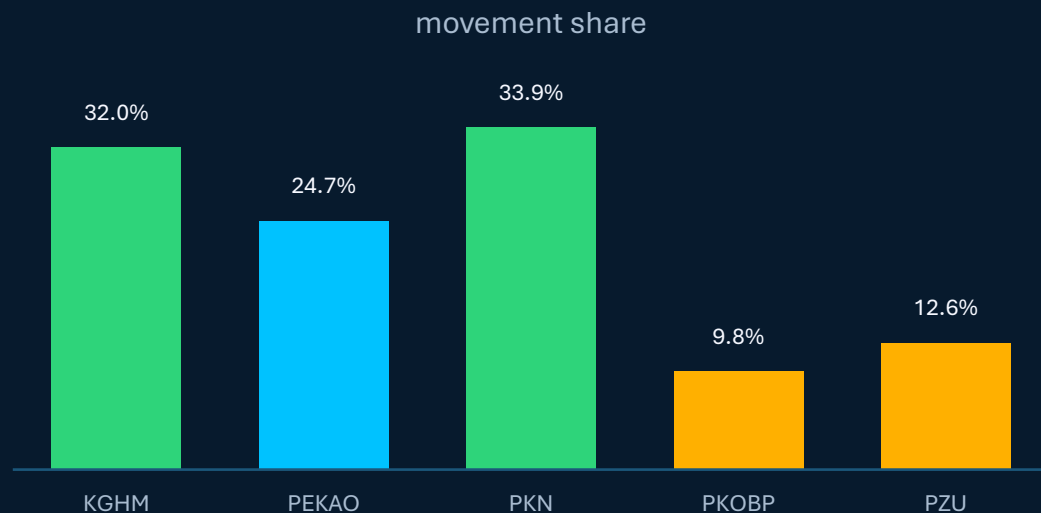
70 / 15 / 15
chronological split



The no-movement class is dominant:

$$Y_t = (0,0)$$

Movement share in the test set:



Some stocks move often enough to learn from.
Others are dominated by no movement.

This will strongly affect how we interpret the models.

$$NLL = -\frac{1}{N} \sum_i \log p_{\theta}(Y_i | X_i)$$

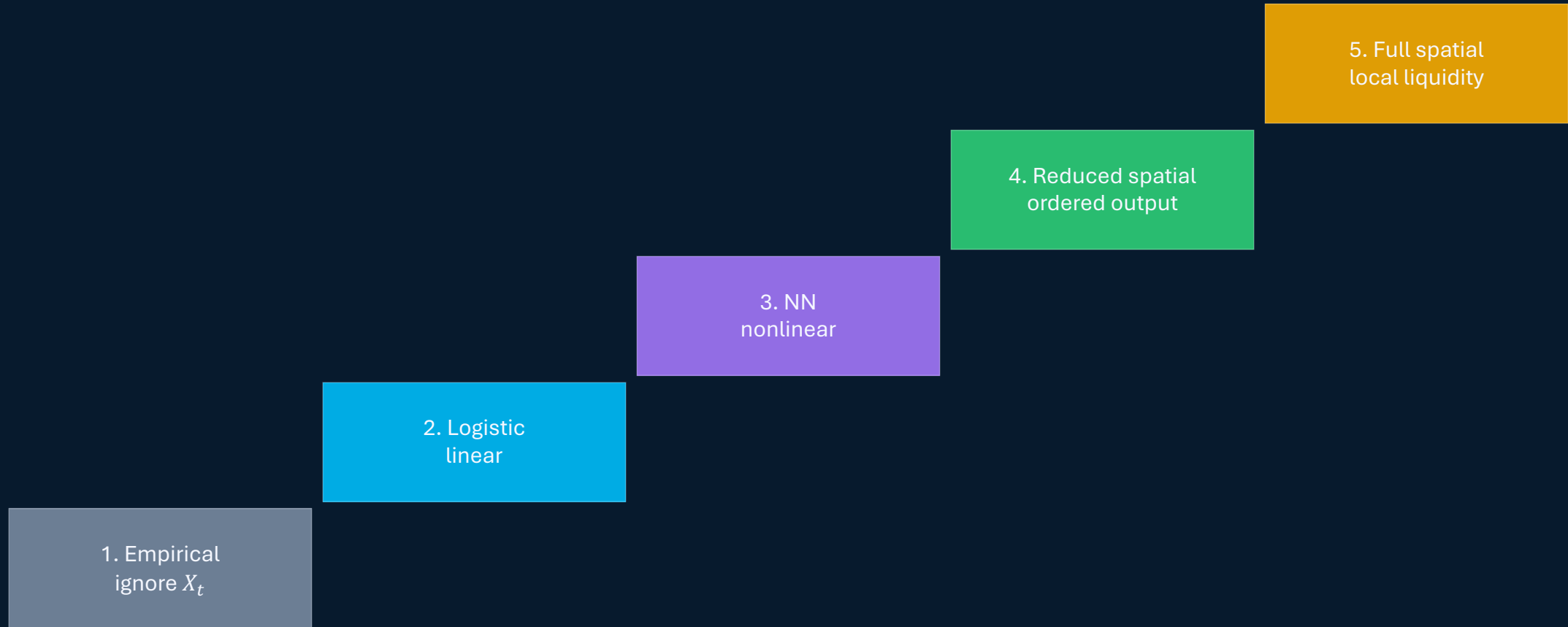
$$MoveNLL = -\frac{1}{|M|} \sum_{i \in M} \log p_{\theta}(Y_i | X_i)$$

$$M = \{i : Y_i \neq (0,0)\}$$

Accuracy is secondary.

A model can be accurate by predicting no movement almost always.

Lower NLL is better.



Each model answers a more specific question about what information is present in the book.

Empirical

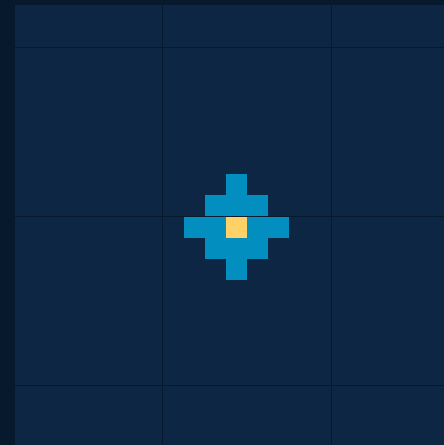
Model 1

The empirical baseline ignores X_t .

$$p_{emp}(y) = \left(\frac{1}{N}\right) \sum_i 1\{Y_i = y\}$$

$$p_{emp}(Y_t = y | X_t) = p_{emp}(y)$$

Purpose:
a minimum benchmark.

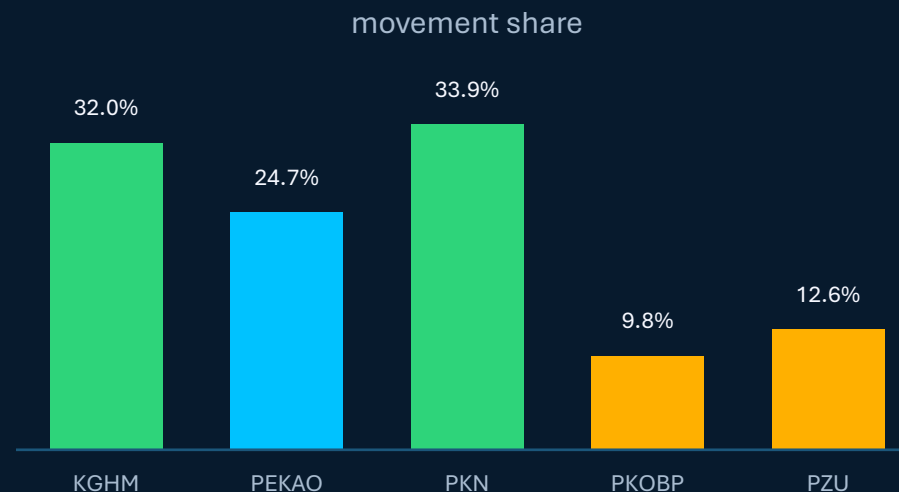


same distribution
for every observation

The empirical model mostly
learns:

"no movement is likely."

Useful reference point, but it cannot use
information in the current book state.



Next question:
Does X_t contain predictive information?

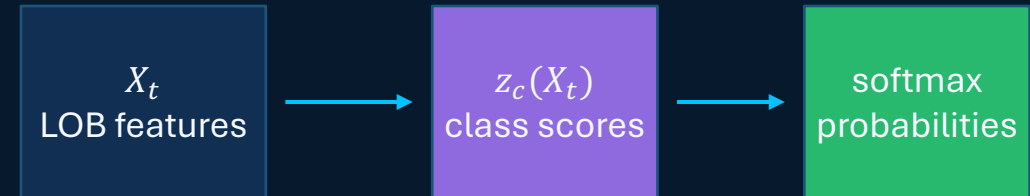
Logistic regression

Model 2

$$z_c(X_t) = \theta_c^T X_t + b_c$$

$$p_{\theta}(Y_t = c | X_t) = \frac{\exp(z_c)}{\sum_j \exp(z_j)}$$

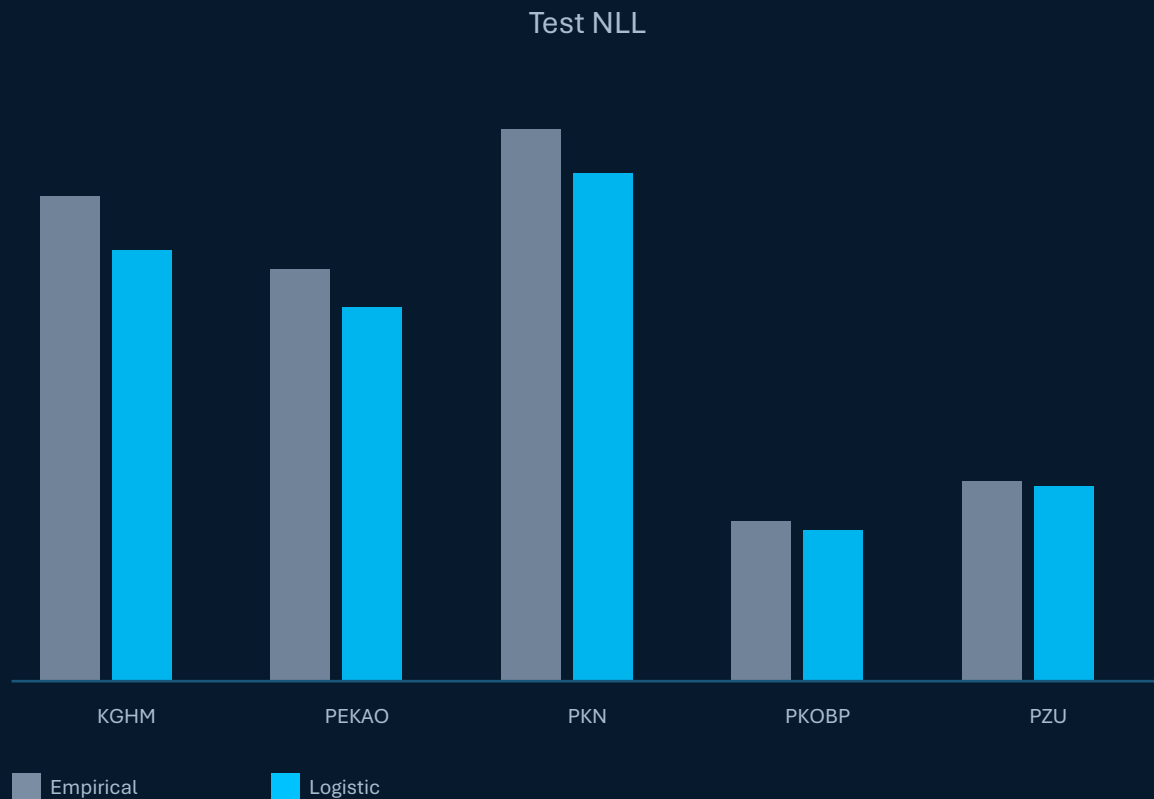
This is a linear conditional model.



Question:
Can a linear model using LOB features
beat the empirical baseline?

Logistic Result

Model 2



Logistic regression improves NLL for every stock.

Interpretation:
 X_t contains useful predictive information beyond unconditional label frequencies.

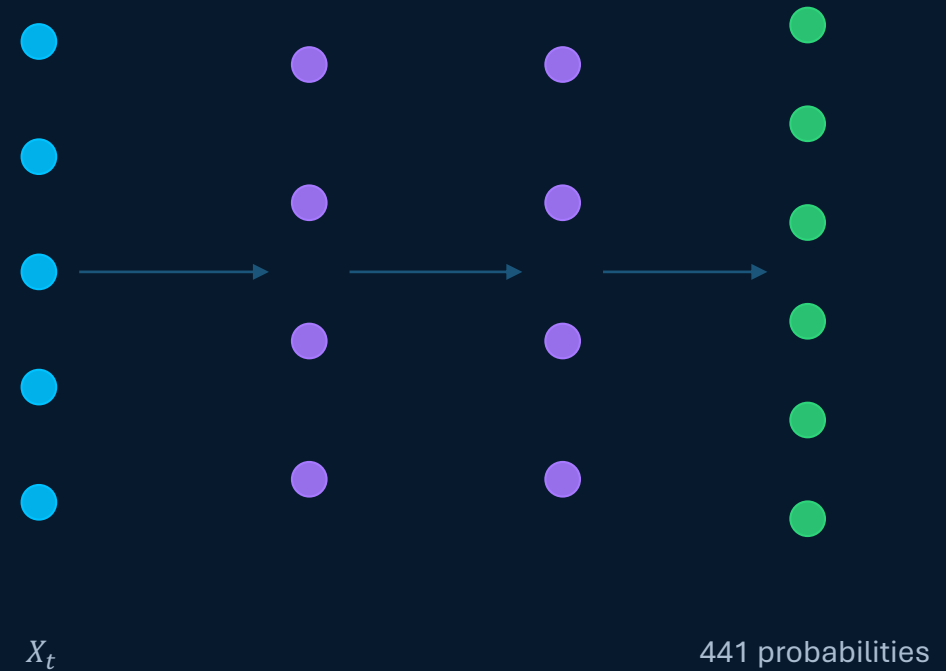
Next: are the relevant relationships nonlinear?

$$h_0 = X_t$$

$$h_m = \varphi(W_m h_{m-1} + b_m)$$

$$p_{\theta}(Y_t = c | X_t) = \text{softmax}(z_{\theta}(X_t))_c$$

Purpose: test whether nonlinear interactions improve prediction.



NN Result

Model 3

Stock	Logistic	Standard NN
KGHM	1.4711	1.3707
PEKAO	1.2788	1.2060
PKNORLEN	1.7369	1.5931
PKOBP	0.5151	0.4755
PZU	0.6665	0.6263

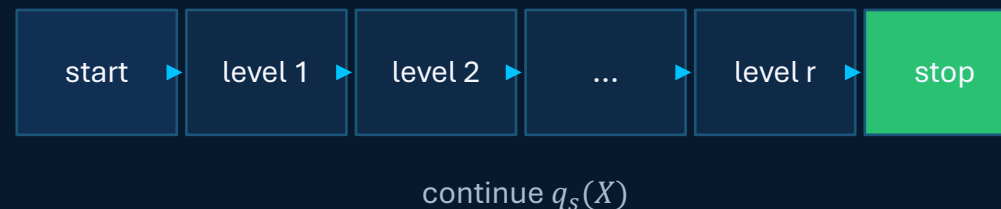
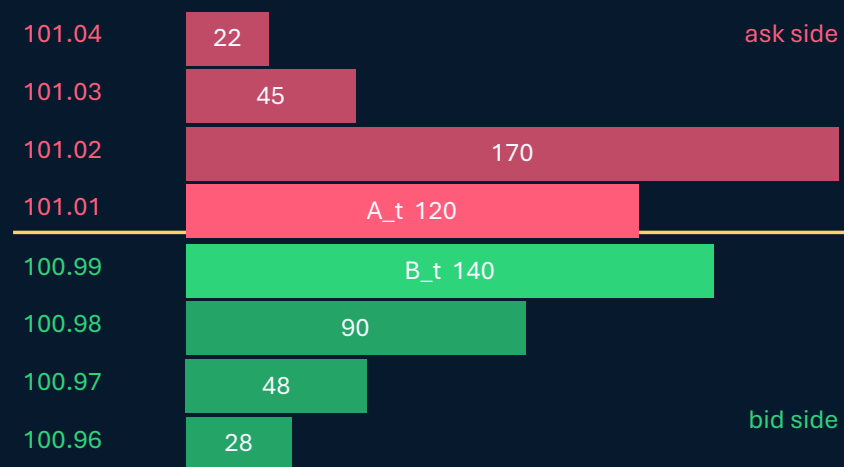
The standard neural network improves over logistic regression.

Interpretation:
nonlinear relationships between depth, spread, imbalance and price levels matter.

Limitation:
441 output classes are treated as unrelated.

Price levels are ordered.

- a 2-tick move must pass through a 1-tick move
- larger movements pass through intermediate levels
- movement probabilities should respect this geometry



This motivates a hazard-style view:
continue through levels until the quote stops.

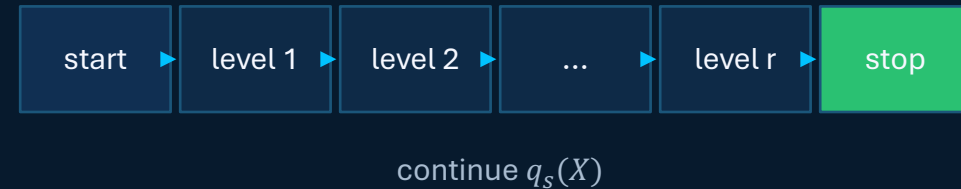
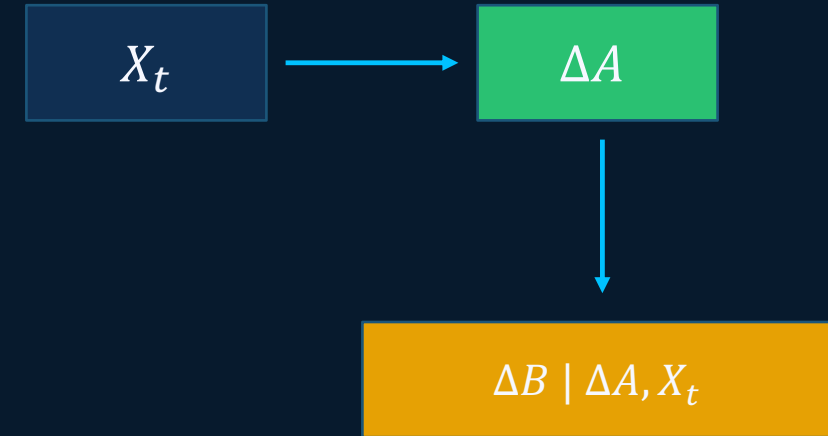
Reduced Spatial

Model 4

$$p(\Delta A, \Delta B \mid X) = p(\Delta A \mid X) p(\Delta B \mid \Delta A, X)$$

- predict ask movement first
- predict bid movement conditionally
- for each side: direction + magnitude
- magnitude is modelled by continuation probabilities

Structured output
without local windows

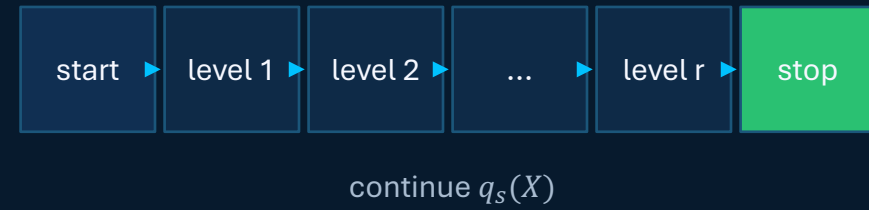


Hazard Form

Model 4

For a positive ask movement $r < K$:

$$p(\Delta A = r | X) = p(\Delta A > 0 | X) [\prod_{s=1}^{r-1} q_s(X)] [1 - q_r(X)]$$



Boundary class:

$$p(\Delta A = K | X) = p(\Delta A > 0 | X) \prod_{s=1}^{K-1} q_s(X)$$

$q_s(X)$ = probability of continuing
beyond level s

Reduced Result

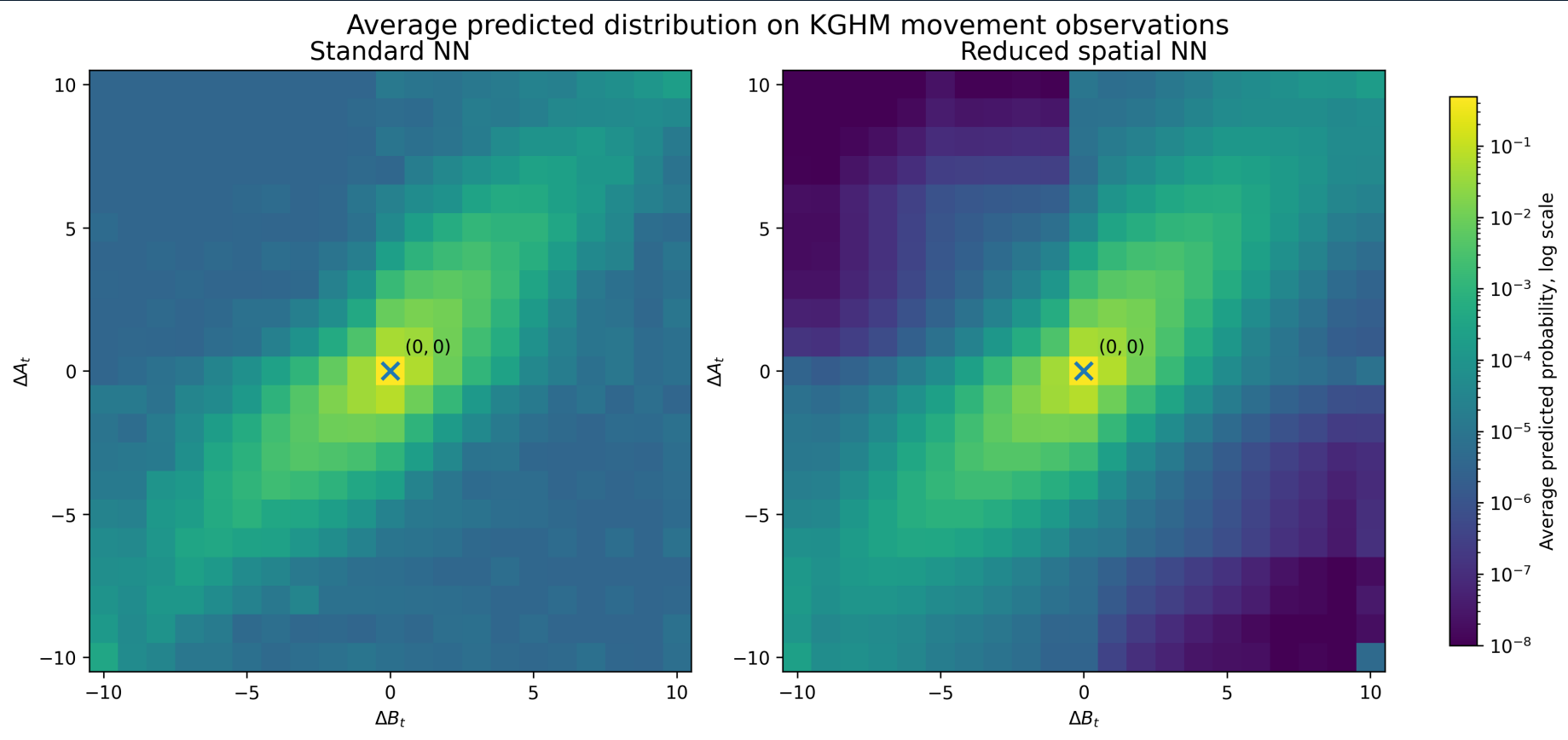
Model 4

Stock	Standard NN	Reduced spatial
KGHM	3.4976	3.4254
PEKAO	3.9864	3.9764
PKNORLEN	3.8439	3.8364
PKOBP	3.8940	3.6885
PZU	3.8172	3.8255

Reduced spatial NN gives best MoveNLL for 4 out of 5 stocks.

Interpretation:
ordered output structure helps assign probability to nonzero quote movements.

This is the strongest support for the spatial idea.



Full Spatial

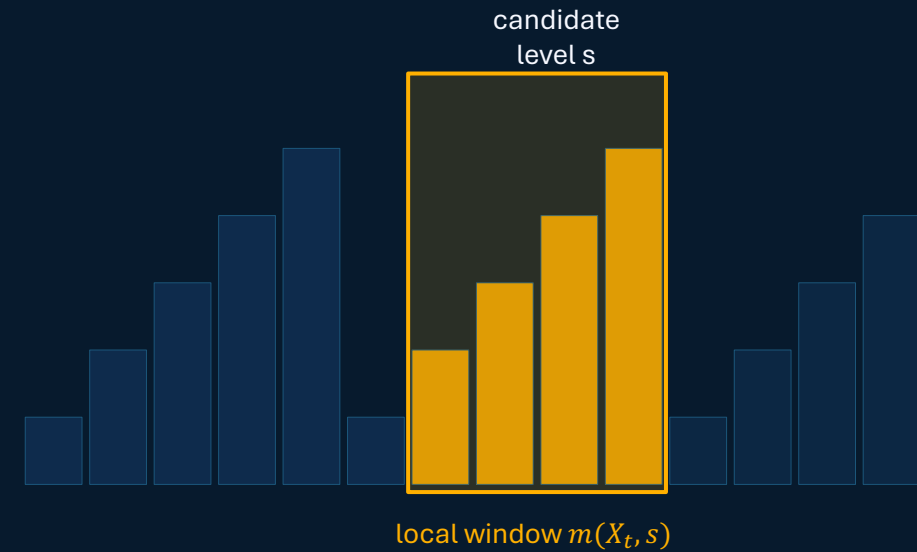
Model 5

The reduced model uses global features.
The full local model uses local liquidity.

$m(X_t, s)$ = local window around level s

$q_s(X_t) = q_\theta(m(X_t, s))$

The same local network is shared across
levels.



local liquidity around candidate level



The two checkpoints expose the trade-off between no-movement calibration and movement prediction.

Stock	Val NLL ckpt	Val MoveNLL ckpt
KGHM	3.5728	3.4184
PEKAO	4.0087	3.8008
PKNORLEN	4.2227	4.1684
PKOBP	4.8688	3.9928
PZU	4.2178	3.8274

Movement-selected checkpoints improve MoveNLL, but often worsen overall NLL.

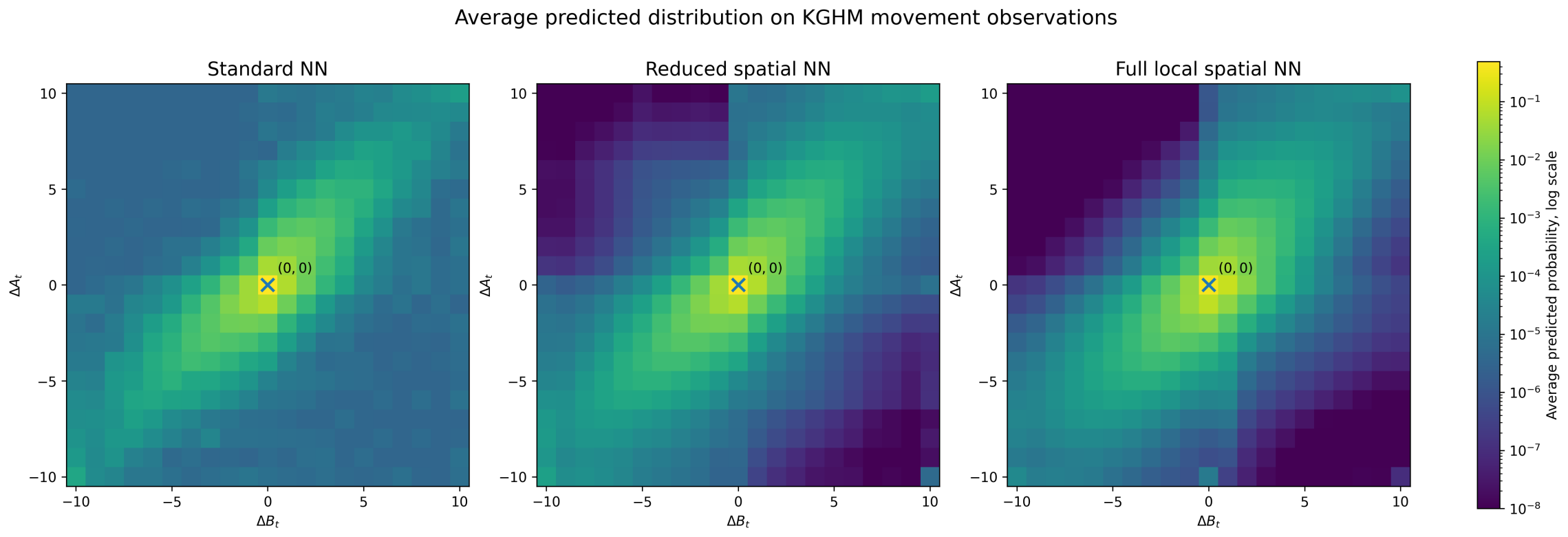
Does full spatial improve over the best main model?

Model 5

Stock	MoveNLL best main	Full spatial	Result
KGHM	3.4254	3.4184	slight improvement
PEKAO	3.9764	3.8008	improvement
PKNORLEN	3.8364	4.1684	worse
PKOBP	3.6885	3.9928	worse
PZU	3.8172	3.8274	roughly equal

Movement-selected checkpoints improve MoveNLL, but often worsen overall NLL.

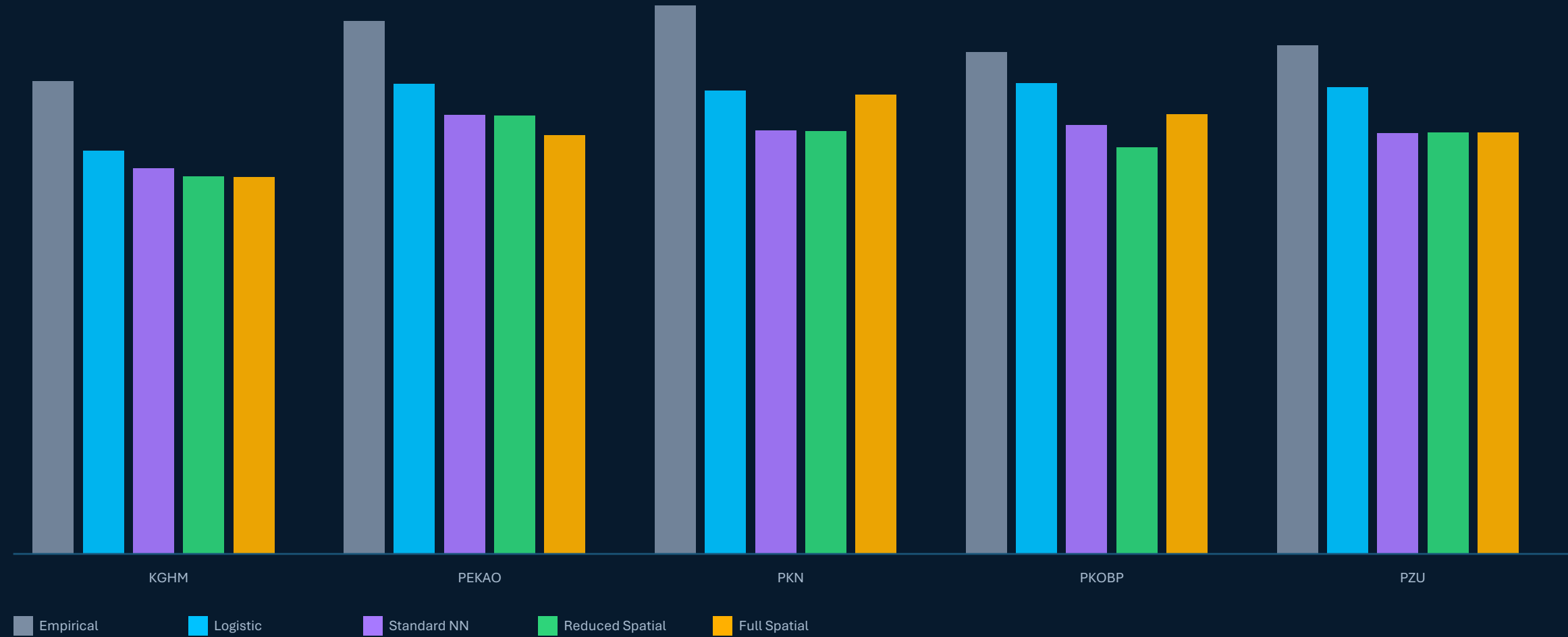
- strongest movement gains: KGHM and PEKAO
- less stable for inactive books
- not a uniform winner



Model	Question	Finding
Empirical	What if we ignore X_t ?	no-movement baseline
Logistic	linear information?	yes
Standard NN	nonlinear information?	yes
Reduced spatial	ordered output?	helps MoveNLL
Full spatial	local liquidity?	mixed, data-dependent

Each step reveals a different type of information in the order book.

Movement-conditional NLL by model and stock



Original paper

Spatial NN strongly dominates
standard NN.

Our replication

Spatial models are competitive, but
do not uniformly dominate.

Key reasons:

The dataset changes the prediction problem
Simplified implementation

Our full model vs the paper

Method

Sirignano (2016)

Our full spatial implementation

 Data	NASDAQ Level III, 489 stocks	WSELOB-2017, 5 reconstructed stocks
 Output space	General spatial distribution	Clipped grid: (Delta A, Delta B) in $[-10,10]^2 = 441$ classes
 Spatial signal	Local map $m(x,y)$ near candidate future prices	Dense tick-grid + local windows around candidate levels
 Architecture	Original spatial neural network	Simplified local hazard-style approximation; shared local network across levels
 Experiment	Large-scale cross-stock study	Single-stock fixed-horizon replication (30 seconds)



Same core idea: use local liquidity to structure the output distribution.



So our full model is closer to the paper, but still not an exact reproduction.

Architecture difference

Method

Same spatial intuition, different implementation.

Sirignano spatial NN

OUTPUT SPACE

General spatial distribution

Designed beyond a fixed 441-class grid

LOCALITY

Local map near candidate future price

$m(x, y)$ selects book information near y

NETWORK STRUCTURE

Direction networks + magnitude networks

h_i for sign, $f_{i,+}, f_{i,-}$ for movement size



adapted

Our full local spatial NN

OUTPUT SPACE

Finite clipped classification grid

$Y = (\Delta A, \Delta B) \in [-10, 10]^2 \rightarrow 441 \text{ classes}$

LOCALITY

Dense tick grid + fixed local windows

$m(X_t, s)$ extracts a window around level s

NETWORK STRUCTURE

Explicit hazard-style continuation

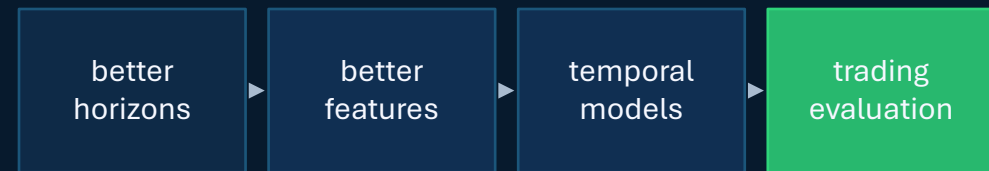
$q_s(X_t) = \sigma(g_\theta(m(X_t, s)))$

Takeaway

We preserve the local-liquidity idea, but replace the paper's general spatial NN by a finite-grid hazard approximation.

- Reduced replication, not exact reproduction
- WSELOB-2017 differs strongly from
- NASDAQ Level III data
- Reconstructed aggregate order book events
- Fixed 30-second horizon for all stocks
- Strong no-movement class imbalance
- Clipped output grid $[-10, 10]^2$
- Simplified dense-grid local map
- Statistical prediction, not trading profitability

- more faithful local map $m(x,y)$, instead of fixed local windows.
- more flexible output space for large quote movements.
- stock-specific or event-time horizons
- order-flow based inputs
- more active datasets
- online updating
- link predictions to execution or trading decisions



1. The current order book state contains predictive information.
2. Neural networks improve upon empirical and linear baselines.
3. Spatial structure helps most clearly on nonzero quote movements.
4. The strength of this conclusion depends on dataset, activity level and metric.

Spatial structure helps, but data determines how much.

Questions

Questions?

GitHub:
https://github.com/ldsRehorst/wi4224_deep_learning_for_limit_order_books.git

